Molar Concentration - Practice worksheet

GCE Advanced Level Chemistry - Concentration molaire

Complete each exercise on paper, writing every step of your reasoning, then compare with the separate answer key.
Fais chaque exercice sur une feuille en detaillant ton raisonnement, puis compare avec le corrige separe.

1. Exercise 1 - From grams to moles

Calculate the number of moles in 10 g of calcium carbonate, CaCO3 (M = 100 g/mol).
Calcule le nombre de moles dans 10 g de carbonate de calcium, CaCO3 (M = 100 g/mol).

2. Exercise 2 - Concentration from moles

0.25 mol of solute is dissolved to make 0.50 L of solution. Compute the molar concentration.
On dissout 0,25 mol de solute pour obtenir 0,50 L de solution. Calcule la concentration molaire.

3. Exercise 3 - The full chain

5.85 g of sodium chloride, NaCl (M = 58.5 g/mol), are made up to 0.50 L of solution. Find the molarity. (First grams to moles, then divide.)
On dissout 5,85 g de chlorure de sodium, NaCl (M = 58.5 g/mol), pour obtenir 0,50 L de solution. Trouve la molarite. (D'abord les grammes en moles, puis on divise.)

4. Exercise 4 - Dilution keeps the moles

10 mL of a 2 mol/L stock solution is diluted to 100 mL. What is the final concentration?
On dilue 10 mL d'une solution mere a 2 mol/L jusqu'a 100 mL. Quelle est la concentration finale ?

5. Exercise 5 - How much mass to weigh

What mass of glucose (M = 180 g/mol) is needed to prepare 0.250 L of a 0.20 mol/L glucose solution?
Quelle masse de glucose (M = 180 g/mol) faut-il peser pour preparer 0,250 L d'une solution de glucose a 0,20 mol/L ?

6. Exercise 6 - Read the bottle

A bottle is labelled 0.50 mol/L. How many moles of solute are there in 250 mL taken from it? Watch the volume unit.
Un flacon porte l'etiquette 0,50 mol/L. Combien de moles de solute y a-t-il dans 250 mL preleves ? Attention a l'unite de volume.

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